Solutions to 2011 Power Question

1a.

$$\begin{array}{lcll}
\mathrm{Pa}(1,1)+\mathrm{Pa}(2,1)+\mathrm{Pa}(3,1)+\mathrm{Pa}(4,1)+\mathrm{Pa}(5,1) & = & 1+2+3+4+5 & = \mathbf{15} \\
\mathrm{Pa}(2,2)+\mathrm{Pa}(3,2)+\mathrm{Pa}(4,2)+\mathrm{Pa}(5,2)+\mathrm{Pa}(6,2) & = & 1+3+6+10+15 & = \mathbf{35} \\
\mathrm{Pa}(3,3)+\mathrm{Pa}(4,3)+\mathrm{Pa}(5,3)+\mathrm{Pa}(6,3)+\mathrm{Pa}(7,3) & = & 1+4+10+20+35 & = \mathbf{70} \\
\mathrm{Pa}(4,4)+\mathrm{Pa}(5,4)+\mathrm{Pa}(6,4)+\mathrm{Pa}(7,4)+\mathrm{Pa}(8,4) & = & 1+5+15+35+70 & = \mathbf{126}.
\end{array}$$

1b. Notice that $\mathrm{Pa}(n,n)+\mathrm{Pa}(n+1,n)+\cdots+\mathrm{Pa}(n+k,n)=\mathrm{Pa}(n+k+1,n+1)$, so $m=n+k+1$ and $j=n+1$. (By symmetry, $j=k$ is also correct.) The equation is true for all n when $k=0$, because the sum is simply $\mathrm{Pa}(n,n)$ and the right side is $\mathrm{Pa}(n+1,n+1)$, both of which are 1. Proceed by induction on k. If $\mathrm{Pa}(n,n)+\mathrm{Pa}(n+1,n)+\cdots+\mathrm{Pa}(n+k,n)=\mathrm{Pa}(n+k+1,n+1)$, then adding $\mathrm{Pa}(n+k+1,n)$ to both sides yields $\mathrm{Pa}(n+k+1,n)+\mathrm{Pa}(n+k+1,n+1)=\mathrm{Pa}(n+k+2,n+1)$ by the recursive rule for Pa.

2a. By definition of Pa, $\mathrm{Pa}(n,0)=\mathrm{Pa}(n,n)=1$ for all nonnegative integers n, and this value is odd, so $\mathrm{PaP}(n,0)=\mathrm{PaP}(n,n)=1$ by definition.

2b.

						1								
					1		1							
				1		0		1						
			1		1		1		1					
		1		0		0		0		1				
	1		1		0		0		1		1			
1		0		1		0		1		0		1		
	1		1		1		1		1		1		1	
1		0		0		0		0		0		0		1

3a. Notice that

$$\begin{aligned}
\mathrm{Pa}(n,k) &= \frac{n!}{k!(n-k)!} \\
&= \frac{n}{k}\frac{(n-1)!}{(k-1)!((n-1)-(k-1))!} \\
&= \frac{n}{k}\cdot\mathrm{Pa}(n-1,k-1).
\end{aligned}$$

Examining the right hand side in the case where $n=2^j$ and $0<k<n$, the second factor, $\mathrm{Pa}(n-1,k-1)$ is an integer, and the first factor has an even numerator, so $\mathrm{Pa}(n,k)$ is even. Therefore $\mathrm{PaP}(n,k)=0$.

3b. Proceed by induction on j. It is easier to follow the logic when the computations are expressed modulo 2. The claim is that $\mathrm{Pa}(n,k)\equiv\mathrm{Pa}(n-2^j,k-2^j)+\mathrm{Pa}(n-2^j,k) \bmod 2$. If $j=0$, so that

$2^j = 1$, then this congruence is exactly the recursive definition of $\mathrm{Pa}(n,k)$ when $0 < k < n$. For $k = 0$ and $k = n$, the left-hand side is 1 and the right-hand side is either $0 + 1$ or $1 + 0$. For other values of k, all three terms are zero.

Now, assume $\mathrm{Pa}(n,k) \equiv \mathrm{Pa}(n-2^j, k-2^j) + \mathrm{Pa}(n-2^j, k) \bmod 2$ for some $j \geq 0$, and let $n \geq 2^{j+1}$. For any k, apply the inductive hypothesis three times:

$$\begin{aligned}\mathrm{Pa}(n,k) &\equiv \mathrm{Pa}(n-2^j, k-2^j) + \mathrm{Pa}(n-2^j, k) \bmod 2 \\ &\equiv \mathrm{Pa}(n-2^j-2^j, k-2^j-2^j) + \mathrm{Pa}(n-2^j-2^j, k-2^j) \\ &\qquad + \mathrm{Pa}(n-2^j-2^j, k-2^j) + \mathrm{Pa}(n-2^j-2^j, k).\end{aligned}$$

But $u + u \equiv 0 \bmod 2$ for any integer u. Thus the two occurrences of $\mathrm{Pa}(n-2^j-2^j, k-2^j)$ cancel each other out, and

$$\begin{aligned}\mathrm{Pa}(n,k) &\equiv \mathrm{Pa}(n-2^j-2^j, k-2^j-2^j) + \mathrm{Pa}(n-2^j-2^j, k) \\ &\equiv \mathrm{Pa}(n-2^{j+1}, k-2^{j+1}) + \mathrm{Pa}(n-2^{j+1}, k) \bmod 2.\end{aligned}$$

3c. By part (b), $\mathrm{Pa}(n,k) \equiv \mathrm{Pa}(n-2^j, k-2^j) + \mathrm{Pa}(n-2^j, k) \bmod 2$ when $j \geq 0$ and $n \geq 2^j$. If $2^j \leq n < 2^{j+1}$ and $0 \leq k < 2^j$, then $\mathrm{Pa}(n-2^j, k-2^j) = 0$, so

$$\mathrm{Pa}(n,k) \equiv \mathrm{Pa}(n-2^j, k-2^j) + \mathrm{Pa}(n-2^j, k) \equiv \mathrm{Pa}(n-2^j, k) \bmod 2.$$

Thus $\mathrm{PaP}(n,k) = \mathrm{PaP}(n-2^j, k)$.

Alternate Solution: Problem 3a establishes the statement when $n = 2^j$. For $2^j < n < 2^{j+1}$, proceed by induction on n. Then $\mathrm{PaP}(n,k) \equiv \mathrm{PaP}(n-1,k-1) + \mathrm{PaP}(n-1,k) \bmod 2$, while $\mathrm{PaP}(n-1,k-1) = \mathrm{PaP}(n-1-2^j, k-1)$ and $\mathrm{PaP}(n-1,k) = \mathrm{PaP}(n-1-2^j, k)$. But $\mathrm{PaP}(n-1-2^j, k-1) + \mathrm{PaP}(n-1-2^j, k) \equiv \mathrm{PaP}(n-2^j, k) \bmod 2$, establishing the statement.

4a.

			24		37		27		9		1	
		30		61		64		36		10		1
	36		91		125		100		46		11		1

4b. Using the given values yields the system of equations below.

$$\begin{cases} \mathrm{Cl}(1,1) = 1 = a(1)^2 + b(1) + c \\ \mathrm{Cl}(2,1) = 7 = a(2)^2 + b(2) + c \\ \mathrm{Cl}(3,1) = 19 = a(3)^2 + b(3) + c \end{cases}$$

Solving this system, $a = 3, b = -3, c = 1$.

4c. Use induction on n. For $n = 1, 2, 3$, the values above demonstrate the theorem. If $\mathrm{Cl}(n,1) = 3n^2 - 3n + 1$, then $\mathrm{Cl}(n+1,1) = \mathrm{Cl}(n,0) + \mathrm{Cl}(n,1) = 6n + (3n^2 - 3n + 1) = (3n^2 + 6n + 3) - (3n + 3) + 1 = 3(n+1)^2 - 3(n+1) + 1$.

5a. $\mathrm{Cl}(11,2) = 1000$.

5b. $\mathrm{Cl}(n,2) = (n-1)^3$. Use induction on n. First, rewrite $\mathrm{Cl}(n,1) = 3n^2 - 3n + 1 = n^3 - (n-1)^3$, and notice that $\mathrm{Cl}(2,2) = 1 = (2-1)^3$. Then if $\mathrm{Cl}(n,2) = (n-1)^3$, using the recursive definition, $\mathrm{Cl}(n+1,2) = \mathrm{Cl}(n,2) + \mathrm{Cl}(n,1) = (n-1)^3 + (n^3 - (n-1)^3) = n^3$.

5c. $\mathrm{Cl}(11,3) = 2025$.

5d. Notice that $\mathrm{Cl}(3,3) = 1 = 1^3$, and then for $n > 3$, $\mathrm{Cl}(n,3) = \mathrm{Cl}(n-1,2) + \mathrm{Cl}(n-1,3)$; replacing $\mathrm{Cl}(n-1,3)$ analogously on the right side yields the summation

$$\mathrm{Cl}(n,3) = \mathrm{Cl}(n-1,2) + \mathrm{Cl}(n-2,2) + \ldots + \mathrm{Cl}(3,2) + 1.$$

By 5b, $\mathrm{Cl}(n-1,2) = (n-2)^3$, so this formula is equivalent to

$$\mathrm{Cl}(n,3) = (n-2)^3 + (n-3)^3 + \ldots + 2^3 + 1.$$

Use the identity $1^3 + 2^3 + \cdots + m^3 = \frac{m^2(m+1)^2}{4}$ and substitute $n-2$ for m to obtain $\mathrm{Cl}(n,3) = \frac{(n-2)^2(n-1)^2}{4}$.

6. Notice that $\mathrm{Cl}(n,0) = 6 \cdot \mathrm{Pa}(n,1)$, and that for $n > 0$, $\mathrm{Cl}(n,n) = \mathrm{Pa}(n,n)$. From problem 4c, $\mathrm{Cl}(n,1) = 6\mathrm{Pa}(n,2) + 1$ (where $\mathrm{Pa}(n,k) = 0$ if $k > n$).

For $k > 1$, repeated application of the formula $\mathrm{Cl}(m,k) = \mathrm{Cl}(m-1,k-1) + \mathrm{Cl}(m-1,k)$ allows each value of $\mathrm{Cl}(n,k)$ to be written as a sum:

$$\begin{aligned}
\mathrm{Cl}(n,k) &= \mathrm{Cl}(n-1,k-1) + \mathrm{Cl}(n-1,k) \\
&= \mathrm{Cl}(n-1,k-1) + \mathrm{Cl}(n-2,k-1) + \mathrm{Cl}(n-2,k), \text{ and eventually:} \\
&= \mathrm{Cl}(n-1,k-1) + \mathrm{Cl}(n-2,k-1) + \cdots + \mathrm{Cl}(k,k-1) + \mathrm{Cl}(k,k) \\
&= \mathrm{Cl}(n-1,k-1) + \mathrm{Cl}(n-2,k-1) + \cdots + \mathrm{Cl}(k-1,k-1),
\end{aligned}$$

because $\mathrm{Cl}(k,k) = \mathrm{Cl}(k-1,k-1) = 1$. Then

$$\begin{aligned}
\mathrm{Cl}(n,2) &= \mathrm{Cl}(n-1,1) + \mathrm{Cl}(n-2,1) + \cdots + \mathrm{Cl}(1,1) \\
&= (6\mathrm{Pa}(n-1,2) + 1) + (6\mathrm{Pa}(n-2,2) + 1) + \cdots + (6\mathrm{Pa}(2,2) + 1) + 1 \\
&= 6(\mathrm{Pa}(n-1,2) + \mathrm{Pa}(n-2,2) + \cdots + \mathrm{Pa}(2,2)) + (n-1).
\end{aligned}$$

By the identity from 1b, $\mathrm{Pa}(n-1,2) + \mathrm{Pa}(n-2,2) + \cdots + \mathrm{Pa}(2,2) = \mathrm{Pa}(n,3)$. Therefore $\mathrm{Cl}(n,2) = 6\mathrm{Pa}(n,3) + n - 1 = 6\mathrm{Pa}(n,3) + \mathrm{Pa}(n-1,1)$.

The general formula is $\mathrm{Cl}(n,k) = 6\mathrm{Pa}(n,k+1) + \mathrm{Pa}(n-1,k-1)$. This formula follows by induction on n. If $n = k$, then $\mathrm{Cl}(n,k) = 1$, and $6\mathrm{Pa}(n,k+1) + \mathrm{Pa}(n-1,k-1) = 6 \cdot 0 + 1 = 1$. Then suppose for some $n \geq k$, $\mathrm{Cl}(n,k) = 6\mathrm{Pa}(n,k+1) + \mathrm{Pa}(n-1,k-1)$. It follows that

$$\begin{aligned}
\mathrm{Cl}(n+1,k) &= \mathrm{Cl}(n,k-1) + \mathrm{Cl}(n,k) \\
&= (6\mathrm{Pa}(n,k) + \mathrm{Pa}(n-1,k-2)) + (6\mathrm{Pa}(n,k+1) + \mathrm{Pa}(n-1,k-1)) \\
&= 6(\mathrm{Pa}(n,k) + \mathrm{Pa}(n,k+1)) + (\mathrm{Pa}(n-1,k-2) + \mathrm{Pa}(n-1,k-1)) \\
&= 6\mathrm{Pa}(n+1,k+1) + \mathrm{Pa}(n,k-1).
\end{aligned}$$

7a.

$$\begin{array}{ccccccccccccc} & & & & \frac{1}{5} & & & & \frac{1}{20} & & & & \frac{1}{5} \\ & & \frac{1}{6} & & & \frac{1}{30} & & \frac{1}{60} & & \frac{1}{60} & & \frac{1}{30} & & \frac{1}{6} \end{array}$$

7b. $\mathrm{Le}(17,1) = \mathrm{Le}(16,0) - \mathrm{Le}(17,0) = \frac{1}{17} - \frac{1}{18} = \frac{1}{306}$.

7c. $\mathrm{Le}(17,2) = \mathrm{Le}(16,1) - \mathrm{Le}(17,1) = \mathrm{Le}(15,0) - \mathrm{Le}(16,0) - \mathrm{Le}(17,1) = \frac{1}{2448}$.

8a. $\mathrm{Le}(n,1) = \mathrm{Le}(n-1,0) - \mathrm{Le}(n,0) = \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$.

8b. Because $\mathrm{Le}(n,1) = \frac{1}{n} - \frac{1}{n+1}$,

$$\begin{aligned} \sum_{i=1}^{2011} \mathrm{Le}(i,1) &= \sum_{i=1}^{2011} \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{2010} - \frac{1}{2011}\right) + \left(\frac{1}{2011} - \frac{1}{2012}\right) \\ &= 1 - \frac{1}{2012} \\ &= \frac{2011}{2012}. \end{aligned}$$

8c. $\mathrm{Le}(n,2) = \mathrm{Le}(n-1,1) - \mathrm{Le}(n,1) = \frac{1}{n(n-1)} - \frac{1}{n(n+1)} = \frac{2}{(n-1)(n)(n+1)}$. Note that this result appears in the table as a unit fraction because at least one of the integers $n-1, n, n+1$ is even.

9a. Extending the result of 8b gives

$$\sum_{i=1}^{n} \mathrm{Le}(i,1) = \frac{1}{1} - \frac{1}{n},$$

so as $n \to \infty$, $\sum_{i=1}^{n} \mathrm{Le}(i,1) \to 1$. This value appears as $\mathrm{Le}(0,0)$, so $n = k = 0$.

9b. $n = k = m - 1$.

9c. Because in general $\mathrm{Le}(i,m) = \mathrm{Le}(i-1,m-1) - \mathrm{Le}(i,m-1)$, a partial sum can be rewritten as follows:

$$\begin{aligned} \sum_{i=m}^{n} \mathrm{Le}(i,m) &= \sum_{i=m}^{n} (\mathrm{Le}(i-1,m-1) - \mathrm{Le}(i,m-1)) \\ &= (\mathrm{Le}(m-1,m-1) - \mathrm{Le}(m,m-1)) + (\mathrm{Le}(m,m-1) - \mathrm{Le}(m+1,m-1)) + \\ &\quad \cdots + (\mathrm{Le}(n-1,m-1) - \mathrm{Le}(n,m-1)) \\ &= \mathrm{Le}(m-1,m-1) - \mathrm{Le}(n,m-1). \end{aligned}$$

Because the values of $\mathrm{Le}(n,m-1)$ get arbitrarily small as n increases (proof: $\mathrm{Le}(i,j) < \mathrm{Le}(i-1,j-1)$ by construction, so $\mathrm{Le}(n,m-1) < \mathrm{Le}(n-m+1,0) = \frac{1}{n-m+1}$), the limit of these partial sums is $\mathrm{Le}(m-1,m-1)$. So $n = k = m-1$.

Note: This result can be extended even further. In fact, for every value of $k < n$,

$$\mathrm{Le}(n,k) = \sum_{i=n+1}^{\infty} \mathrm{Le}(i,k+1).$$

In other words, each entry in Leibniz's triangle is an infinite sum of the entries in the diagonal directly to its right, beginning with the entry below and to the right of the given one.

10. Note that $\frac{1}{3} = \mathrm{Le}(2,0) = \mathrm{Le}(3,0) + \mathrm{Le}(3,1)$. Also $\mathrm{Le}(3,0) = \mathrm{Le}(4,0) + \mathrm{Le}(4,1)$ and $\mathrm{Le}(4,0) = \mathrm{Le}(5,0) + \mathrm{Le}(5,1)$. So

$$\begin{aligned} \frac{1}{3} &= \mathrm{Le}(5,0) + \mathrm{Le}(5,1) + \mathrm{Le}(4,1) + \mathrm{Le}(3,1) \\ &= \frac{1}{6} + \frac{1}{30} + \frac{1}{20} + \frac{1}{12}. \end{aligned}$$

Therefore $a = 6, b = 12, c = 20, d = 30$ is a solution.

On the other hand, a similar analysis yields

$$\begin{aligned} \frac{1}{3} &= \mathrm{Le}(4,0) + \mathrm{Le}(5,1) + \mathrm{Le}(5,2) + \mathrm{Le}(3,1) \\ &= \frac{1}{5} + \frac{1}{30} + \frac{1}{60} + \frac{1}{12}, \end{aligned}$$

so $a = 5, b = 12, c = 30, d = 60$ is another such solution.

Finally, notice that $\mathrm{Le}(3,1) = \frac{1}{12} = \mathrm{Le}(11,0)$, so that $\frac{1}{12}$ can be rewritten as $\mathrm{Le}(12,0) + \mathrm{Le}(12,1) = \frac{1}{13} + \frac{1}{156}$. Then

$$\begin{aligned} \frac{1}{3} &= \mathrm{Le}(4,0) + \mathrm{Le}(4,1) + \mathrm{Le}(3,1) \\ &= \mathrm{Le}(4,0) + \mathrm{Le}(4,1) + \mathrm{Le}(12,0) + \mathrm{Le}(12,1) \\ &= \frac{1}{5} + \frac{1}{20} + \frac{1}{13} + \frac{1}{156}, \end{aligned}$$

yields $a = 5, b = 13, c = 20, d = 156$.

In general, once a triple a, b, c has been found, rewriting (for example) $\frac{1}{a} = \mathrm{Le}(a-1,0) = \mathrm{Le}(a,0) + \mathrm{Le}(a,1) = \frac{1}{a+1} + \frac{1}{a(a+1)}$ creates an appropriate quadruple, although on occasion these values duplicate another value in the quadruple.

11. The formula is $\mathrm{Le}(n,k) = \frac{1}{(n+1)\cdot \mathrm{Pa}(n,k)}$, or equivalently, $\frac{1}{(k+1)\cdot \mathrm{Pa}(n+1,k+1)}$. Because $\mathrm{Pa}(n,0) = \mathrm{Pa}(n,n) = 1$, when $k = 0$ or $k = n$, the formula is equivalent to the definition of $\mathrm{Le}(n,0) = \mathrm{Le}(n,n) = \frac{1}{n+1}$.

To prove the formula for $1 \le k \le n-1$, use induction on k. The base case $k = 0$ was proved above. If the formula holds for a particular value of $k < n$, then it can be extended to the case $k+1$ using two identities:

$$\begin{aligned} \mathrm{Le}(n+1,k+1) &= \mathrm{Le}(n,k) - \mathrm{Le}(n+1,k) \text{ and} \\ (n+1)\mathrm{Pa}(n,k) &= \frac{(n+1)!}{k!(n-k)!}. \end{aligned}$$

Using the first identity and the inductive hypothesis yields

$$\begin{aligned} \mathrm{Le}(n+1,k+1) &= \mathrm{Le}(n,k) - \mathrm{Le}(n+1,k) \\ &= \frac{1}{(n+1)\cdot \mathrm{Pa}(n,k)} - \frac{1}{(n+2)\cdot \mathrm{Pa}(n+1,k)}. \end{aligned}$$

Applying the second identity to the right side of the equation yields:

$$\begin{aligned} \mathrm{Le}(n+1,k+1) &= \frac{k!\,(n-k)!}{(n+1)!} - \frac{k!(n+1-k)!}{(n+2)!} \\ &= \frac{k!\,(n-k)!}{(n+2)!}\left[(n+2)-(n+1-k)\right] \\ &= \frac{k!\,(n-k)!}{(n+2)!}\,(k+1) \\ &= \frac{(k+1)!\,((n+1)-(k+1))!}{(n+1)!\,(n+2)} \\ &= \frac{1}{(n+2)\mathrm{Pa}(n+1,k+1)}. \end{aligned}$$